Niloy Das

Nationality: Bangladeshi | **Phone:** (+880) 1974337950 (Mobile) | **Email address:** dasnil684@gmail.com | **LinkedIn:** Niloy Das | **ResearchGate:** ResearchGate | **Address:** 10/4 West Kanda Para, Narsingdi Sadar, Narsingdi, 1600, Narsingdi, Bangladesh (Home)

ABOUT MYSELF

An independent researcher specializing in Quantum Machine Learning (QML) and hybrid quantum-classical architectures, with a focus on scalable solutions beyond the limits of classical AI. Research interests span quantum-enhanced optimization for next-generation wireless systems (ISAC, LEO satellites) and large-scale intelligent systems. Additional expertise in Explainable AI (XAI), with emphasis on developing clinically interpretable diagnostic tools that are both high-performing and trustworthy.

EDUCATION & TRAINING

01/01/2020 - 18/08/2025 - NOAKHALI, BANGLADESH

B.SC. (ENGINEERING) IN INFORMATION AND COMMUNICATION ENGINEERING- NOAKHALI SCIENCE AND TECHNOLOGY UNIVERSITY

Completed a four-year undergraduate degree in Information and Communication Engineering, graduating with a CGPA of 3.90 out of 4.00 and ranking first in my graduating class. My coursework included digital communications, computer programming, computer fundamentals, architectures, signal processing, machine learning, artificial intelligence, and computer networks.

Field(s) of study: Information and Communication Technologies | **Final grade:** First Class (CGPA 3.90/4.00) | **Level in EQF:** 6 | **Number of credits:** 171 | **Thesis:** EMADNet (Enhanced Multi-Scale Adaptive Denoising Network)

01/06/2017 - 31/07/2019 - NARSINGDI, BANGLADESH

HIGHER SECONDARY CERTIFICATE (HSC) - NARSINGDI IMPERIAL COLLEGE, NARSINGDI

Completed higher secondary education in science, specializing in Physics, Chemistry, Mathematics, and Biology, which built a strong foundation in quantitative, analytical, and problem-solving skills.

Field(s) of study: Science | **Final grade:** 5.00 | **Level in EQF:** 4

01/01/2012 - 31/05/2017 - NARSINGDI, BANGLADESH

SECONDARY SCHOOL CERTIFICATE (SSC) - BRAHMONDI KAMINI KISHOR MOULIK GOVERNMENT HIGH SCHOOL, NARSINGDI

Maintained strong and consistent academic performance while developing leadership, teamwork,communication, and adaptability through organizing academic events.

Field(s) of study: Science | **Final grade:** 5.00 | **Level in EQF:** 3

20/11/2024 - 04/12/2024 - NOAKHALI , BANGLADESH

INDUSTRIAL TRAINING- BANGLADESH TELECOMMUNICATIONS COMPANY LIMITED

- Completed 15-day industrial training at Bangladesh Telecommunications Company Limited (BTCL), a state-owned national telecommunications operator
- Gained hands-on exposure to telecommunications infrastructure and network operations in a professional government institution
- Observed fiber optic and transmission systems used in national backbone telecommunications networks
- Trained under the supervision of the Department of ICE, NSTU, as part of the undergraduate curriculum requirements

WORK EXPERIENCE

01/02/2024 - 30/04/2024 - NOAKHALI , BANGLADESH

SOFTWARE DEVELOPMENT INTERN BUSINESS AUTOMATION LIMITED

- Developed web applications using PHP (Laravel), HTML, CSS, and JavaScript within a professional team environment
- Built and integrated Machine Learning models using Python into web-based solutions
- Performed API testing with Postman and managed relational databases using MySQL

01/01/2023 - CURRENT - NOAKHALI , BANGLADESH

PRIVATE TUTOR SELF-EMPLOYED

- Provided one-on-one academic tutoring to SSC and HSC level students in Science subjects
- Developed customized lesson plans and conducted regular assessments to track student progress
- Demonstrated strong communication and teaching skills in a self-managed, freelance role

SKILLS

Programming & Development
Python | PHP | C++ | MATLAB | Web Frameworks: (.NET ,Laravel ,Django,Angular) | Web Technologies: HTML5, CSS3 | Databases (MongoDB MySql PostgresSql)

Artificial Intelligence
Machine Learning • | Computer VIsion | Artificial İntelligence | Deep Learning frameworks: PyTorch,Tensorflow,Keras

Tools & Platforms
LaTeX / Microsoft Office | Microsoft Office (Word Excel Powerpoint) | Git/Github, Docker, Gitlab

Research & Academic Skills
Scientific Writing (Scientific Paper Publication) | Data Gathering, Data Visualisation, Data Analysis | Research Methodology writing

Leadership & Teamwork
Event Organization | Team management & team work

LANGUAGE SKILLS

Mother tongue(s): **BENGALI**

	UNDERSTANDING		SPEAKING		WRITING
	Listening	Reading	Spoken production	Spoken interaction	
ENGLISH	C2	C2	C1	C1	C1

PROJECTS

15/12/2024 - 15/05/2025

EMADNet: Enhanced Multi-Scale Adaptive Denoising Network for Image Restoration *Supervised by Tanvir Zaman Khan, Assistant Professor, Department of Information and Communication Engineering, NSTU.*

- Developed a novel CNN architecture for multi-noise image restoration combining multi-scale feature extraction, adaptive filtering, and dual attention mechanisms.
- Implemented in PyTorch, achieving PSNR 40+ dB across Gaussian, salt-and-pepper, and speckle noise types.

04/04/2024 - 20/04/2024

Library Management System The final internship project at Business Automation Limited. [Link]

- Collaborated in team to build a full-stack Library Management System using Laravel (PHP), MySQL, and RESTful APIs to manage library resources, inventory, and member workflows.

PUBLICATIONS

Enlightened Prognosis: Hepatitis Prediction with an Explainable Machine Learning Approach 2025

DAS, N., Hossain, M. B., Adhikary, A., Raha, A. D., Qiao, Y., Hassan, M. M., & Bairagi, A. K. (2025). Enlightened prognosis: Hepatitis prediction with an explainable machine learning approach. PloS ONE, 20(4), e0319078. PLOS ONE

OTFS-Enabled ISAC Scheduling for LEO Satellite with Reconfigurable Holographic Surfaces Using... 2026

Hassan, S. S., Adhikary, A., DAS, N., Sharma, S., & Ratnarajah, T. (2026). OTFS-Enabled ISAC Scheduling for LEO Satellite with Reconfigurable Holographic Surfaces Using a Transformer Approach. IEEE WCNC 2026

PSO-XAI: An Optimized and Interpretable Approach for Breast Cancer Diagnosis 2026

Raquib, M., DAS, N., Prity, F. S., Fahim, A. A., Murad, S. A., Hossain, M. A., ... & Moni, M. A. (2025). PSO-XAI: A PSO-Enhanced Explainable AI Framework for Reliable Breast Cancer Detection. arXiv preprint arXiv:2510.20611. Applied Computational Intelligence and Soft Computing

Domain-Aware Hybrid Quantum Learning via Correlation-Guided Circuit Design for Crime Pattern 2026

Submitted to IEEE GLOBECOM 2026

DAS, N., Adhikary, A., Hassan, S. S., Qiao, Y., Han, Z., Ratnarajah, T., & Hong, C. S. (2026). A Novel Edge-Assisted Quantum-Classical Hybrid Framework for Crime Pattern Learning and Classification. arXiv preprint arXiv:2604.07389. IEEE GLOBECOM 2026

A Multi-Method Framework for Depression Detection: Integrating TextBlob and VADER with Clustering and Fuzzy Sentiment Analysis 2026

Submitted to International Conference on PECCII 2026

Muniat, F., Khan, T. Z., DAS, N., Soheli, S. J., Qiao, Y., & Adhikary, A. (2026). A multi-method framework for depression detection: Integrating TextBlob and VADER with clustering and fuzzy sentiment analysis. International Conference on PECCII

Transformer-Based OTFS-ISAC Scheduling in Cooperative LEO RHS Systems with Physics-Aware Learning 2026

Manuscript in preparation.

Hassan, S. S., Adhikary, A., DAS, N., & Ratnarajah, T. (2026). Transformer-based OTFS-ISAC scheduling in cooperative LEO RHS systems with physics-aware learning. Manuscript in Preparation.

Edge-Assisted Quantum-Classical Hybrid Architectures for Domain and Correlation-Aware Crime Pattern Analytics 2026

Manuscript in preparation.

DAS, N., Adhikary, A., Qiao, Y., Gain, M., Ahsan, M. S., Han, Z., & Hong, C. S. (2026). Quantum-inspired hybrid learning framework for edge-enabled crime pattern identification. Manuscript in Preparation.

A Hybrid CNN-Based Transformer Framework for SAR-Based Flood Detection 2026

Manuscript in preparation

DAS, N., Raquib, M., Islam, S., & Rahman, A. (2026). A hybrid CNN-based transformer framework for SAR-based flood detection. Manuscript in Preparation.

CERTIFICATIONS

11/12/2025 IDP Education
IELTS Academic

IELTS Academic: 7.5 (L: 8.5, R: 8.5, W: 7.0, S: 6.5). [Certificate]

30/04/2024 Business Automation Limited
Certificate of Internship (Software Development)

[Certificate]

01/01/2025 Bangladesh Telecommunications Company Limited (BTCL)
Certificate of Completion (Industrial Training)

[Certificate]

HONOURS AND AWARDS

01/05/2025 Noakhali Science and Technology University (NSTU)
University Merit-Based Scholarship Awarded for 8 consecutive semesters for consistently ranking in the top 3 of the class.

11/01/2018 Board of Intermediate and Secondary Education, Dhaka
General Grade Scholarship (SSC) Awarded scholarship in recognition of achieving a GPA of 5.00/5.00 in the Secondary School Certificate examination.

NETWORKS AND MEMBERSHIPS

03/03/2024 - 01/10/2025 ICE Programming Club, Noakhali Science and Technology University
Vice President

VOLUNTEERING

22/08/2024 - 25/09/2024 Noakhali, Bangladesh
Manusher Jonno Manush Foundation Volunteer & Organizer. Contributed across multiple relief activities, including food and water distribution, fundraising, and community support for flood-affected populations.

01/09/2024 - 03/09/2024 Noakhali Science and Technology University
Prothom Alo Bondhushova Tree Plantation Program

RECOMMENDATIONS

Anupam Kumar Bairagi, Ph.D. Co-author and research collaborator

I have known Niloy Das for several years through academic collaboration. He excels in rigorous research, machine learning solutions, and scientific writing. I highly recommend him as a talented and dedicated researcher with great academic potential.

Email anupam@cse.ku.ac.bd

Apurba Adhikary, Ph.D. Research Supervisor, Collaborator and Course Instructor

I have known Niloy Das for several years as his academic supervisor and research collaborator. He excels in tackling complex problems in Explainable AI, Quantum Machine Learning, and next-generation wireless systems with impressive creativity. I highly recommend him as a promising young researcher.

Email apurba@nstu.edu.bd